UNIVERSITÄT
LEIPZIG

Asymmetric Numeral System

December 18, 2024

Huo Jiang

Fakultät für Mathematik und Informatik
Universität Leipzig

Outline

1. Introduction
2. Entropy Coding
   - 2.1 Huffman Coding
   - 2.2 Arithmetic Coding
   - 2.3 Asymmetric Numeral System
3. Asymmetric Numeral System
   - 3.1 Basic Idea
     - 3.1.1 Motivation
     - 3.1.2 Symbol spread function
     - 3.1.3 Representation of information
     - 3.1.4 Symmetric or Asymmetric
   - 3.2 rANS
     - 3.2.1 Key Concepts
     - 3.2.2 Encoding Formula
   - 3.3 tANS
   - 3.4 Performance
4. Summary

Introduction

Introduction

- Hi there, this is a self inspired latex beamer template for Leipzig university.

Entropy Coding

1st frame

- This is one of the section[1].
- Let's check for 2nd footnote [2].
- The footnotes will be added from the bottom. If you want to pull the footnotes towards red-black margin line then use `\addfootspace{}` at the end of the `\frame{}`.
- `\addfootspace{}` actually adds an additional vacant footnote (without counting that vacant footnote).
- To add more foot space add more `\addfootspace{}`.

[1] This is a long footnote. Here I am. Who am I?

[2] This is a 2nd long footnote. Here I am. Who am I?

2nd frame

- Here is the next section[3]
- Here I used two `\addfootspace{}`. Don't do `\addfootspace{\vspace{1cm}}`.

[3] This is a 2nd long footnote. Here I am. Who am I?

Asymmetric Numeral System

Motivation

ANS combines the compression ratio of arithmetic coding (which uses a nearly accurate probability distribution), with a processing cost similar to that of Huffman coding [3]

Basic Idea

- Symbol spread function

Basic Idea

- Symbol spread function
- Representation of information
  (represents an entire message as a single number, similar to Arithmetic Coding) –> example: representation of {2,0,2,5,1,8}

Basic Idea

- Symbol spread function
- Representation of information
  (represents an entire message as a single number, similar to Arithmetic Coding) –> example: representation of {2,0,2,5,1,8}
- Symmetric or Asymmetric

Basic Idea

- Symbol spread function
- Representation of information
  (represents an entire message as a single number, similar to Arithmetic Coding) –> example: representation of {2,0,2,5,1,8}
- Symmetric or Asymmetric
- rANS and tANS

Symbol spread function

- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]

Symbol spread function

- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]
- $\bar{s} : \mathbb{N} \rightarrow \mathcal{A}, \bar{s}(x) = \text{mod}(x, b) = s$

Symbol spread function

- the function that determines the split of the $\mathbb{N}$ subsets corresponding to different symbols [1]
- $\bar{s} : \mathbb{N} \rightarrow \mathcal{A}, \bar{s}(x) = \text{mod}(x, b) = s$
- In the standard binary system example: $\bar{s}(x) = \text{mod}(x, 2)$
  e.g. splits natural numbers into subsets of even/odd numbers [1]

Symbol spread function (Exa. 1)

$S = \{0, 1\}$

Exa. 1: Standard Binary System

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
s=0																			
s=1																			

Fig.1

Symbol spread function (Exa. 2)

$S = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Exa. 2: Standard Decimal System

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
s=0																					
s=1																					
s=2																					
s=3																					
s=4																					
s=5																					
s=6																					
s=7																					
s=8																					
s=9																					

Fig.2

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

$$\mathcal{A} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

$$\mathcal{A} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

- the sequence of symbols/digits to be encoded:

$$S_{in} = (2, 0, 2, 5)$$

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

$$\mathcal{A} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

- the sequence of symbols/digits to be encoded:

$$S_{in} = (2, 0, 2, 5)$$

- How can we represent this sequence as a single number?

Representation of information (Exa. 2)

- a finite sequence of symbols/digits:

$$\mathcal{A} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

- the sequence of symbols/digits to be encoded:

$$S_{in} = (2, 0, 2, 5)$$

- How can we represent this sequence as a single number?
- The easiest way is to concatenate the symbols/digits:

$$X_{out} = 2025$$

Representation of Information (Exa. 2)

- encoding the sequence of digits (2,0,2,5,1,8)

$$
\begin{aligned}
X_{\text{init}} &= 0 \\
X_1 &= X_{\text{init}} \times 10 + 2 = 2 \\
X_2 &= X_1 \times 10 + 0 = 20 \\
X_3 &= X_2 \times 10 + 2 = 202 \\
X_{\text{out}} &= X_3 \times 10 + 5 = 2025
\end{aligned}
$$

Representation of Information (Exa. 2)

– What is the encoding function?

Representation of Information (Exa. 2)

- What is the encoding function?
- Encoding function

$$X_{i+1} = X_i \times 10 + s_i$$

Representation of Information (Exa. 2)

- What is the encoding function?
- Encoding function

$$X_{i+1} = X_i \times 10 + s_i$$

- x-th appearance of symbol s

$$X' = C(s, X) = X \cdot 10 + s$$

Representation of Information (Exa. 2)

Exa. 2: Standard Decimal System

X' = 10X + s

s=0 s=2 s=2 s=5

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		202		2025
s=0	0										1										2				
s=1		0										1													
s=2			0										1												
s=3				0										1											
s=4					0										1							...		...	
s=5						0										1									
s=6							0										1								
s=7								0										1							
s=8									0										1						
s=9										0										1					

Fig.3

Representation of Information (Exa. 1)

Exa. 1: Standard Binary System

$X' = 2X + s$

s=1 s=0 s=0 s=1

X'	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
X s=0	0		1		2		3		4		5		6		7		8		9
X s=1		0		1		2		3		4		5		6		7		8	

Fig.4

Exa. 1: Standard Binary System

$X' = 2X + s$

s=1 s=0 s=0 s=1

X'	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
x s=0	0		1		2		3		4		5		6		7		8		9
x s=1		0		1		2		3		4		5		6		7		8	

Exa. 2: Standard Decimal System

$X' = 10X + s$

s=2 s=0 s=2 s=5

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		202		2025
s=0	0										1										2				
s=1		0										1													
s=2			0										1												
s=3				0										1											
s=4					0										1							...		...	
s=5						0										1									
s=6							0										1								
s=7								0										1							
s=8									0										1						
s=9										0										1					

Fig.5

Asymmetric Numeral System | Basic Idea | Symmetric or Asymmetric

Exa. 1: Standard Binary System

X' = 2X + s

s=1 s=0 s=0 s=1

X'	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
x s=0	0		1		2		3		4		5		6		7		8		9
x s=1		0		1		2		3		4		5		6		7		8	

Exa. 2: Standard Decimal System

X' = 10X + s

s=2 s=0 s=2 s=5

N	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		202		2025
s=0	0										1										2				
s=1		0										1													
s=2			0										1												
s=3				0										1											
s=4					0										1							...		...	
s=5						0										1									
s=6							0										1								
s=7								0										1							
s=8									0										1						
s=9										0										1					

Fig.6

Symmetric or Asymmetric

- Symmetric: uniform distribution
- Asymmetric: non-uniform distribution

Exa. 3

- Let $\mathcal{A} = \{0, 1\}$
- $Pr(0) = 1/4, Pr(1) = 3/4$
- a cycle of length 4, which contains 1 zero and 3 ones
- $X' = C(s, X) \approx X/p_s$
- $C(0, X) = 4X$, $C(1, X) = 4\lfloor X/3 \rfloor + mod(X, 3) + 1$

rANS

- range
-

Encoding Formula:

$$x' = \left\lfloor \frac{x}{F_s} \right\rfloor \cdot m + \mathrm{mod}(x, F_s) + c_s$$

- **Alphabet set:** $\mathcal{A} = \{a_1, a_2, \ldots, a_k\}$
- **Absolute frequency of symbol** s**:** F_s
- **Total frequency:**

$$m = \sum_{i=1}^{k} F_i$$

- **Probability of symbol** s**:**

$$p_s = \frac{F_s}{m}$$

- **Cumulative frequency before** s**:**

$$c_s := f_0 + f_1 + \cdots + f_{s-1}$$

tANS

Performance

Summary

Summary

- To summarize, this may help you to make a Leipzig university beamer presentation.

Reference

Reference

1. J. Duda, K. Tahboub, N. J. Gadgil and E. J. Delp, ”The use of asymmetric numeral systems as an accurate replacement for Huffman coding,” 2015 Picture Coding Symposium (PCS), Cairns, QLD, Australia, 2015, pp. 65-69, doi: 10.1109/PCS.2015.7170048.
2. 2
3. `https://en.wikipedia.org/wiki/Asymmetric_numeral_systems`